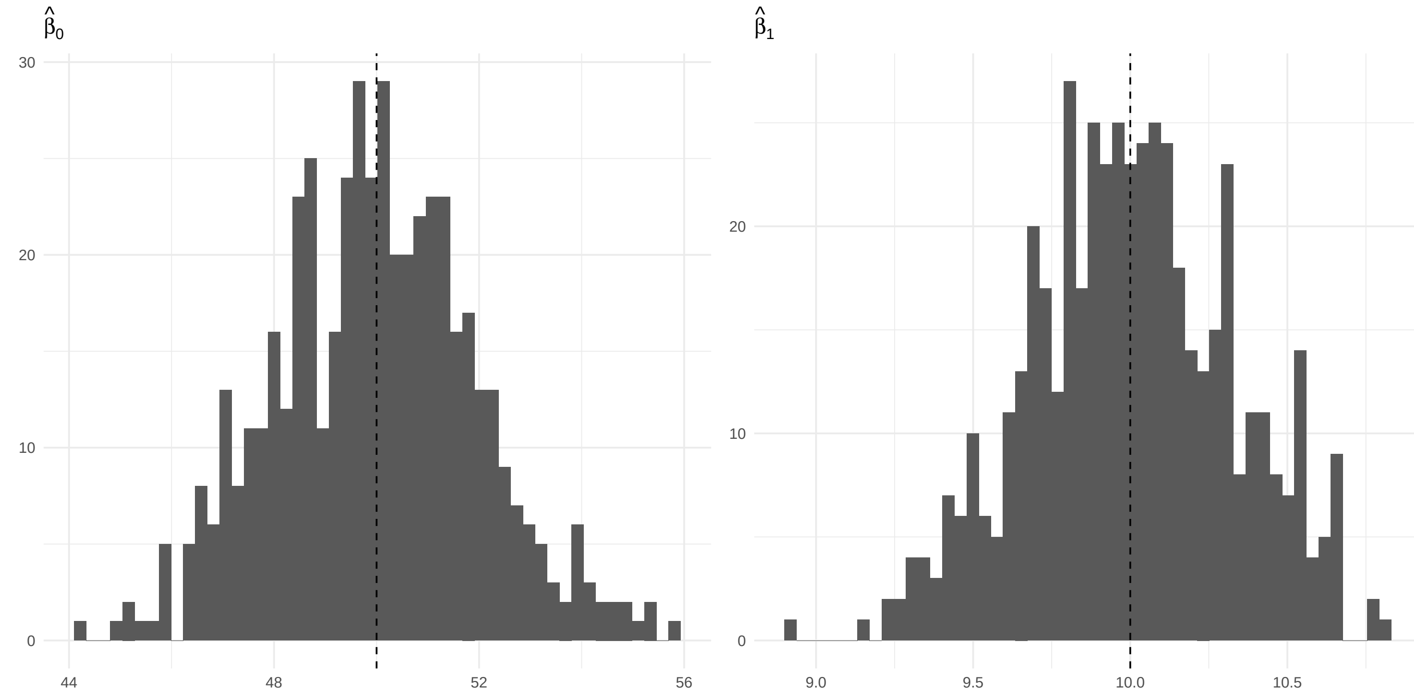
(a) histograms of slope & intercept



DISCUSSION:

The histograms display the sampling distributions of the estimated intercept ($\hat{\beta}_0$) and slope ($\hat{\beta}_1$) from 500 simulated samples generated under the model $y = 50 + 10x + \varepsilon,\ \varepsilon \sim N(0, 16)$, with $n = 20$ observations.

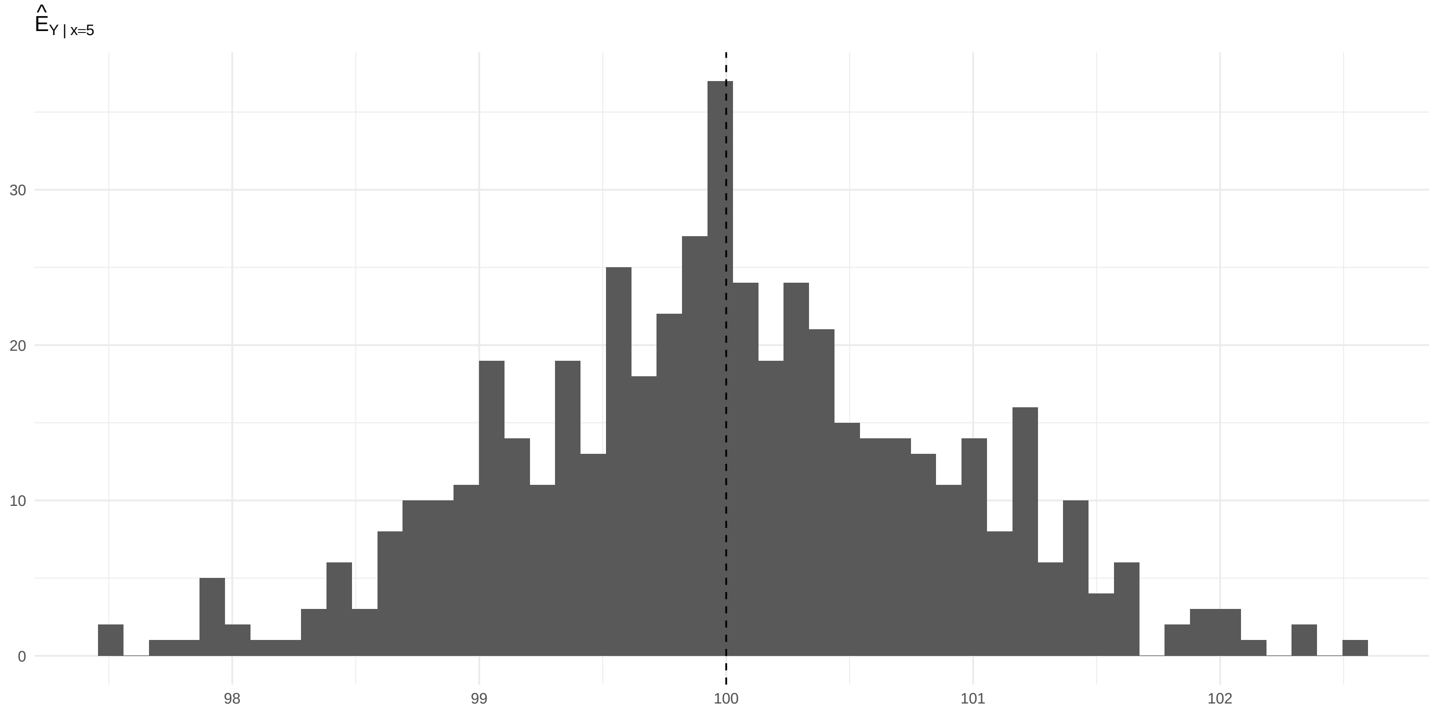
Both distributions appear **bell-shaped and symmetric**, indicating that the least-squares estimators follow an **approximately normal distribution** when the model errors are normally distributed. The dashed vertical lines represent the true parameter values $\beta_0 = 50$ and $\beta_1 = 10$.

The histograms are centered around their true parameter values, confirming that the estimators are **unbiased**. The slope estimator ($\hat{\beta}_1$) shows a narrower spread than the intercept ($\hat{\beta}_0$), reflecting its smaller theoretical variance:

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \quad \mathrm{Var}(\hat{\beta}_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right).$$

Overall, the near-normal shape and centering of the distributions validate the theoretical properties of the OLS estimators—unbiasedness, normality, and predictable variability.

(b) histogram of Ehat[Y|x=5]



DISCUSSION:

The histogram illustrates the sampling distribution of the estimated mean response $\widehat{E}[Y|x=5]$ from 500 simulated samples. Each estimate was computed using $\widehat{E}[Y|x=5] = \hat{\beta}_0 + \hat{\beta}_1(5)$, where $\hat{\beta}_0$ and $\hat{\beta}_1$ are the least-squares estimates from each regression.

The distribution appears **approximately normal, symmetric, and centered near 100**, which is the true mean value at $x = 5$ given by $E[Y|x = 5] = 50 + 10(5) = 100$.
This confirms that the estimator is **unbiased**, with an expected value equal to the true mean response.

The histogram also shows a **narrow spread**, indicating low sampling variability — as expected when x_0 is close to the mean of the predictor values.
No skewness or extreme values are evident, reinforcing the conclusion that $\widehat{E}[Y|x = 5]$ follows an approximately normal distribution under the assumptions of the simple linear regression model.

(c) coverage for β_1

```
## [1] 0.95
```

```
## [1] 475
```

From the 500 simulated samples, **475** of the 95% confidence intervals contained the true slope value $\beta_1 = 10$, giving an empirical coverage proportion of **0.95**.

This means that exactly **95%** of the constructed intervals captured the true slope — which matches the theoretical expectation for a 95% confidence interval. Therefore, the simulation confirms that the OLS-based confidence intervals for β_1 are correctly calibrated and perform as expected under the normal error assumption.

(d) coverage for E[Y|x=5]

```
## [1] 0.952
```

```
## [1] 476
```

From the 500 simulated samples, **476** of the 95% confidence intervals for the mean response $E[Y|x = 5]$ contained the true value of **100**, giving an empirical coverage proportion of **0.952**.

This means that approximately **95.2%** of the constructed intervals captured the true mean response, which aligns closely with the theoretical expectation of 95% coverage.

Hence, the simulation confirms that the confidence intervals for $E[Y|x = 5]$ are well-calibrated and perform as expected under the normal error assumption of the simple linear regression model.